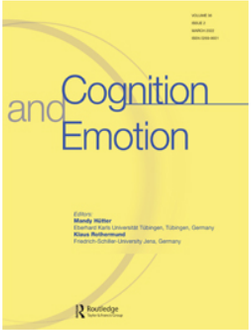

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The emotional adaptation aftereffect discriminates between individuals with high and low levels of depressive symptoms

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ABSTRACT

The adaptation aftereffect plays a critical role in human development and survival. Existing studies have found that, compared with general individuals, individuals with learning disability, autism and dyslexia show a smaller amount of non-affective-based cognitive adaptation aftereffect. Nevertheless, it is unclear whether individuals with depression or depression tendency show similar phenomenon in the adaptation aftereffect, and whether such depression tendency occurs in the non-affective-based cognitive or emotional adaptation aftereffect. To address this question, the present study conducted two experiments. Experiments 1A and 1B used the emotional facial expression adaptation paradigm to examine whether Chinese participants showed the emotional adaptation aftereffect and whether the emotional adaptation aftereffect was influenced by physical features of faces, respectively. Experiment 2 recruited two groups of participants, with high and low depression, respectively, to examine whether they showed differences in the emotional or cognitive adaptation aftereffect. Results showed that Chinese participants showed the typical emotional adaptation aftereffect, which was not influenced by physical features of faces. More importantly, compared to the low-depression group, the high-depression group showed a smaller emotional adaptation aftereffect, but the two groups showed a similar cognitive adaptation aftereffect. These results suggest that level of depressive symptoms is associated with the emotional adaptation aftereffect.

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Adaptation aftereffect; adaptation; depression; emotion; cognition

The adaptation aftereffect plays a critical role in human survival and development (Breton & Sbragia, 2019; Della-Maggiore et al., 2017; Fedorov et al., 2018; Khandan et al., 2018; Lazarus, 1991; Terziev, 2018) because it results in a change or a bias in subsequent perceptual experiences and is conducive to the continuous adaptation of human environment to seek for development. It refers to sensory thresholds that continue to change in response to stimuli, resulting in increased or decreased sensitivity to the subsequent stimuli (Hong & Yoon, 2018; Smit & Wandel, 2006). For instance, after viewing an oriented line for some time, the subsequent lines appear to be tilted away from the adapting orientation (Clifford, 2002; Gibson & Radner, 1937). Adaptation can occur at different levels according to the depth of cognitive processing (Webster, 2011; Webster & MacLeod, 2011). Low-level adaptation has been found for the processing of color, orientation, and direction of motion (Krekelberg et al., 2006; Kuriki, 2007; Thompson & Burr, 2009; Tskhay & Rule, 2015). At a higher cognitive level, adaptation has been reported for

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aspect ratio (Suzuki & Cavanagh, 1998), three-dimensional viewpoint (Fang & He, 2005), surface reflectance (Goddard et al., 2010), and words and objects (Perrachione et al., 2016; Reindl et al., 2018). Such adaptation aftereffects have been linked to higher-level cognitive (dis)abilities. Compared with general individuals, individuals with various learning disabilities such as dyslexia and autistic children show differences in the cognitive adaptation aftereffect (García-Villamisar et al., 2010; Perrachione et al., 2016; Turi et al., 2015). For example, Perrachione et al. (2016) found that the cognitive adaptation aftereffects in response to several types of stimuli such as spoken words, written words and visual objects were significantly smaller in the group with dyslexia than in the control group.

Besides the non-affective-based cognitive adaptation aftereffects mentioned above (referred as cognitive adaptation aftereffect below), in the socio-emotional domain, previous research has further probed the high-level affective-based adaptation aftereffect (i.e. emotional adaption aftereffect) by using the human faces that have been labelled as a typical high-level, socially-relevant category (Furl et al., 2007; Turi et al., 2015). Specifically, emotional adaptation refers to the deviation or change of the individual's emotional perception caused by the influence of the environment or situation (Wang et al., 2017; Webster & MacLeod, 2011; Xu et al., 2008; Yang et al., 2010). In other words, after processing a particular type of emotional information, individuals would show biases or changes in subsequent emotional perception (Burton et al., 2016; Wang et al., 2017; Webster et al., 2004; Xu et al., 2008). Given emotion's importance in species' survival (Khandan et al., 2018; Kiecolt-Glaser et al., 2002; Manning et al., 2019) as well as individuals' life quality (Booth et al., 1998; Fedorov et al., 2018; Terziev, 2018), it is imperative to understand the emotional adaption aftereffect.

Several groups of researchers (Ellamil et al., 2008; Xu et al., 2008; Ying & Xu, 2017; Zamuner et al., 2017) have investigated the mechanisms and influencing factors of the emotional adaptation aftereffect. In those studies, researchers typically used the emotional adaptation paradigm, in which an adapting stimulus and a test stimulus were successively presented. The adapting stimuli were facial images or human voices that carried emotional information. The test stimuli were a series of facial images created by morphing a neutral image with emotional images. Participants were asked to judge whether a test image was happy or sad. They found significant emotional adaptation aftereffect. Specifically, viewing an emotional adapting image shifted individuals' response to the subsequent testing images away from the response in the neutral adaptor condition. Pell and Richards further found that the emotional adaptation aftereffect could be enhanced by using a longer duration of adaptor and by using the adapting and test facial images of the same person (Pell & Richards, 2013). There is also evidence for vocal emotion adaptation aftereffects (Bestelmeyer et al., 2010; Bestelmeyer et al., 2018) and cross-modal emotional adaptation aftereffects (Kitagawa & Ichihara, 2002; Konkle et al., 2009; Pye & Bestelmeyer, 2015; Skuk & Schweinberger, 2013; Wang et al., 2017; Watson et al., 2014). For instance, previous research has found that aftereffects can be observed for adaptation to affective vocal expressions (Bestelmeyer et al., 2010). Moreover, Wang et al. (2017) used voices and faces with different emotions as the adapting stimuli, and asked participants to make judgments on a series of morphed faces after the adapting stimuli. They found that both the unimodal adaptation condition (i.e. both the adapting and the test stimuli were faces) and the cross-modal adaption condition (i.e. the adapting stimuli were voices, but the test stimuli were faces) showed significant emotional adaptation effects.

Although the non-affective-based cognitive adaptation aftereffect has been linked to cognitive (dis)abilities as mentioned earlier, no study thus far has examined whether the emotional adaptation aftereffect is associated with emotional problems (e.g. depression or depression tendency). The current study was designed to test the hypothesis that depression would be associated with the emotional adaptation aftereffect. Our hypothesis was based on previous research that has linked the processing of facial expressions (e.g. recognition accuracy, response biases and attention biases) with depression (Bhagwagar et al., 2004; Gollan et al., 2014; Pringle et al., 2010; Surguladze et al., 2004), although there is also evidence that major depression is related to cognitive impairments (Gallassi et al., 2001; Rock et al., 2014; Thoma & Daum, 2006). Specifically, previous studies found that individuals with depression showed an increased tendency to interpret ambiguous, neutral, or other emotional expressions negatively (Bourke et al., 2010; Surguladze et al., 2004; Suslow et al., 2004).

Two experiments were included in this study. Experiment 1A aimed to confirm the emotional adaptation aftereffect in a Chinese sample. Although several previous studies have found the emotional adaption aftereffect in different populations such as Singaporeans and Americans (Cohen Kadosh et al., 2010; Hong & Yoon, 2018; Pye & Bestelmeyer, 2015; Skuk & Schweinberger, 2013; Xu et al., 2012), no study has investigated it with Chinese participants. Therefore, empirical confirmation was needed. In addition, previous studies typically used adapting stimuli and test stimuli from the same face provider to explore emotional adaptation aftereffects (Wang et al., 2017; Xu et al., 2008; Ying & Xu, 2017). Experiment 1B used adapting stimuli and test stimuli from different face providers to further investigate whether the emotional adaptation effect was caused by individuals' high level of emotional adaptation or by the adaptation to the facial physical features. In Experiment 2, we recruited participants with low and high levels of depression to examine their emotional adaptation aftereffect. They completed the emotional adaptation task used in Experiment 1 as well as a non-emotional cognitive adaptation task as a comparison, to further examine whether the between-group differences in adaptation aftereffect were emotion-specific.

Experiment 1A

To confirm the emotional adaptation aftereffect in Chinese participants, participants were instructed to judge the emotion of a presented facial image after adapting to a happy, neutral, or sad facial image. We expected to replicate the typical emotional adaptation aftereffect (Webster et al., 2004).

Methods

Participants

We recruited 20 young college students (12 females, mean age = 20.6, SD = 2.34) for this study. The number of participants was determined based on the following two aspects. First, following the methods in Uenod et al. (2018), we used Google Scholar to retrieve 30 studies on the adaptation aftereffect which published from January 2015 to December 2020. Among those 30, 9 articles used within-subject designs are directly related to this research. A random-effects meta-analysis of the 9 articles resulted in an integrated effect size of Cohen's $d = 1.12$ (95% CI [0.83, 1.41]). Power analysis on this integrated effect size revealed that a sample size of 20 would give power of more than 0.88. Second, it is generally accepted that the ideal statistical test power and effect size should be higher than 0.80 (Cohen, 1988). To detect a medium effect size (i.e. 0.60) with 0.80 power in a one-sample *t*-test, the expected sample size is 19 participants. Therefore, the number of participants was enough for the investigation of adaptation aftereffect. All participants were strongly right-handed (Snyder & Harris, 1993), had normal or corrected-to-normal vision and reported no history of neurological or psychiatric disorders. Before the experiment, all participants provided written informed consent, and this study was approved by the Institutional Review Board of School of Psychology at South China Normal University.

Materials

Following previous studies on emotional adaptation (Wang et al., 2017; Webster et al., 2004; Xu et al., 2008; Ying & Xu, 2017), we used three adapting facial images and eight test images as materials. They were facial images of one male person taken from the Karolinska Directed Emotional Faces database (Lundqvist et al., 1998). The adapting images consisted of three facial images (happy, sad and neutral). These images were then used to generate test images by using WebMorph (STOIK Imaging, https://webmorph.org). Specifically, following previous studies (Wang et al., 2017; Ying & Xu, 2017), we morphed the sad image with the neutral image to generate a series of images with the proportion of happiness varying from 0 (saddest) to 0.5 (neutral) and morphed the neutral image with the happy image to generate a series of images with the proportion of happiness varying from 0.5 (neutral) to 1.0 (happiest). Eight images with proportions of happiness equal to 0.15, 0.3, 0.4, 0.5, 0.6, 0.7, 0.85 and 1.0 were chosen as test images (Figure 1A).

Procedures

Experiment 1A included three adapting conditions: the happy, sad and neutral adaptor conditions, with the happy, sad and neutral facial image as the adapting stimulus, respectively. Each condition was performed in a separate run with the order of conditions counterbalanced across participants. There was a 5-min break after each run to avoid the carryover of the aftereffects. In each run, 80 trials

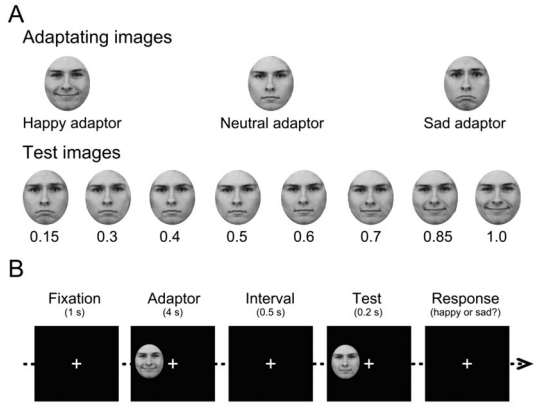

Figure 1. Stimuli and experimental design. (A) Three types of adaptors (i.e. happy adaptor, neutral adaptor and sad adaptor) were followed by a series of morphed test images from sad to happy. (B) Each adapting stimulus was presented for 4 s, followed by a 0.5-s inter-stimuli interval. The test image was then presented for 0.2 s. Participants were instructed to judge whether a test image was happy or sad by pressing one of two buttons.

were randomly presented, with 10 repetitions for each test image. Each trial began with a 1-s fixation cross, followed by an adapting image for 4 s. After a 0.5-s inter-stimulus interval, a test image was presented for 0.2 s (Figure 1B). The relatively short duration was used for the test stimulus because the duration of 0.2 s has been repeatedly used and validated by previous studies (Hong & Yoon, 2018; Wang et al., 2017; Webster et al., 2004; Xu et al., 2008; Ying & Xu, 2017) and because there was evidence that the adaptation aftereffect is more pronounced for the duration of 0.2–0.3 s (Wolfe, 1984; Xu et al., 2008). Participants had to judge whether the test face was happy or sad by pressing one of two keys (press an "f" key for happy, press a "j" key for sad) as accurately and quickly as possible. The key assignment was counterbalanced across participants during the experiment. Because previous studies have found that the adaptation aftereffect is stronger when the stimuli are presented in the visual periphery than at the fovea (Luo et al., 2015; Wang et al., 2017), all facial stimuli were presented on the left side of the central fixation cross, and participants were asked to gaze at the fixation in the center of the screen. This experiment lasted for 40 min. The study follows exactly the same procedures used in two previous studies (Wang et al., 2017; Ying & Xu, 2017). The experimental data collection was compiled using the Psychtoolbox (www.psychtoolbox.org) in MATLAB (www.mathworks.com).

Data analysis

Data analysis was performed using IBM SPSS Statistics 22 (IBM SPSS Statistics V22.0). We first computed the fraction of "happy" responses to each test image. Then, we fitted the psychometric function for the fraction of "happy" responses as a function of the proportion of happiness of the test face for each participant. The function was a sigmoidal function in the form of $f(x) = 1/(1 + e^{-a(x-b)})$, where b was the test-stimulus parameter corresponding to 50% of the psychometric function (i.e. the point of subjective equality, PSE), and a/4 was the slope of the function at the PSE. Next, we calculated the magnitude of the adaptation aftereffect (i.e. PSE-shift) by subtracting the PSE of the baseline condition (neutral adaptor) from that of an emotional adaptation condition (i.e. happy adaptor or sad adaptor) for each participant. Finally, the significance of the adaptation aftereffect for the two emotional adaption conditions was examined by using one-sample t-tests.

Results and discussion

The fraction of happiness responses and reaction times in the three conditions were shown in Table S1. We used a sigmoidal function to fit the response for each participant. The parameter of model fit (i.e. R^2) was 0.94 ± 0.15 in the happy adaptor condition, 0.98 ± 0.03 in the neutral adaptor condition, and 0.81 ± 0.43 in sad adaptor condition. These results suggest that the function fitted the pattern of responses well for all participants.

As shown in Figure 2, both the happy adaptor (mean PSE-shift = 0.12) and the sad adaptor (mean PSE-shift = −0.1) produced significant adaptation aftereffects. Specifically, for the happy adaptor, PSE-shift was significantly higher than 0 (t (19) = 5.69, $p < .001$, Cohen's $d = 1.23$), suggesting that more sad judgments to the test images were made in the happy adaptor condition relative to the baseline condition. For the sad adaptor, PSE-shift was significantly lower than 0 (t (19) = −5.35, $p < .001$, Cohen's $d = -1.12$), suggesting that more happy judgments to the test images were made in the sad adaptor condition relative to the baseline condition.

The results of Experiment 1A are consistent with previous behavioural findings of emotional adaptation aftereffect (Wang et al., 2017; Webster et al., 2004; Xu et al., 2008; Ying & Xu, 2017). However, together with previous studies (Wang et al., 2017;

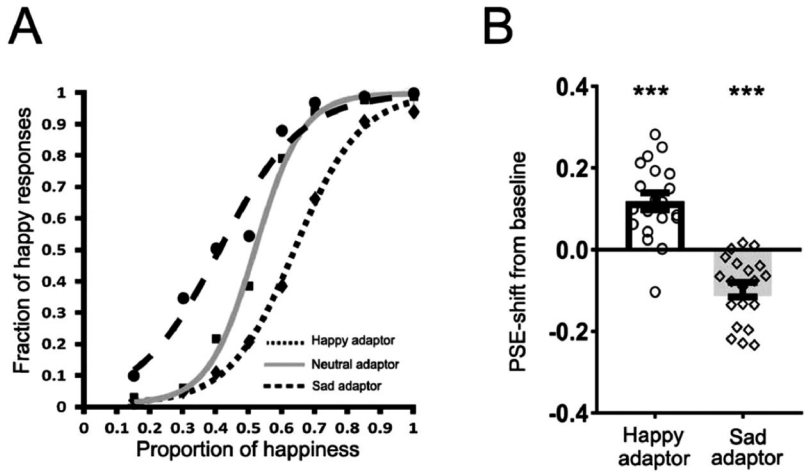

Figure 2. The emotional adaptation aftereffect in Experiment 1A. (A) The fraction of happy responses is plotted as a function of the proportion of happiness of the test images, separately for the happy adaptor condition (R^2: 0.94 ± 0.15), the neutral adaptor condition (R^2: 0.98 ± 0.03), and the sad adaptor condition (R^2: 0.81 ± 0.43); (B) The mean PSE-shift from the baseline condition (i.e. the neutral adaptor condition) for all participants. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. Error bars represent standard errors.

Xu et al., 2008; Ying & Xu, 2017), this experiment used the adapting stimuli and test stimuli from the same face provider to explore emotional adaptation aftereffects. Therefore, it is possible that the emotional adaptation aftereffect might be caused by the similarity in physical features between the adapting and test images. To exclude that possibility, Experiment 1B used the adapting images and test images as materials to replicate the results in Experiment 1A.

Experiment 1B

Experiment 1B used the adapting images and test images from different face providers to replicate the emotional adaptation aftereffect found in Experiment 1A. We expected that the emotional adaptation aftereffect would not be influenced by the facial features.

Methods

Participants

We recruited 20 young college students (12 females, mean age = 20, SD = 1.22) for this experiment. All participants were strongly right-handed (Snyder & Harris, 1993), had normal or corrected-to-normal vision and reported no history of neurological or psychiatric disorders. Before the experiment, all participants provided written informed consent, and this experiment was approved by the Institutional Review Board of School of Psychology at South China Normal University.

Materials

As in Experiment 1A, the materials also consisted of 3 adapting facial images and 8 test facial images (Figure 3). The adapting and test images were respectively acquired from two male faces taken from the same face database as in Experiment 1A. The morphing procedure for the test images was the same as in Experiment 1A.

Procedures and data analysis

The same as in Experiment 1A.

Results and discussion

The behavioural responses in the three conditions were shown in Table S2. As in Experiment 1A, we fitted the response for each participant. The parameter of model fit (i.e. R^2) was 0.96 ± 0.08 in the happy adaptor condition, 0.96 ± 0.04 in the neutral adaptor condition, and 0.89 ± 0.29 in sad adaptor condition. These results suggest that the function fitted the pattern of responses well for all participants.

We then examined the emotional adaptation aftereffects. As shown in Figure 4, both the happy adaptor (mean PSE-shift = 0.12) and the sad adaptor (mean PSE-shift = −0.1) produced significant adaptation aftereffects. Specifically, for the happy adaptor, the PSE-shift was significantly higher than 0 (t (19) = 2.14, $p < .05$, Cohen's $d = 0.48$), which showed that the judgment of the face test was sadder than the

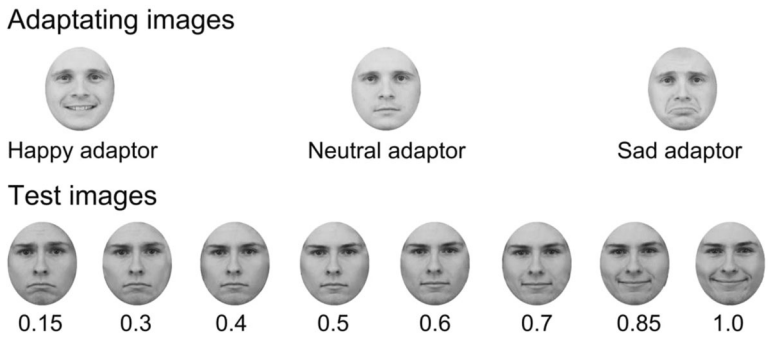

Figure 3. Experimental stimuli in Experiment 1B. Three types of adaptors (i.e. happy adaptor, neutral adaptor and sad adaptor) were followed by a series of morphed test images from sad to happy.

baseline condition. In contrast, for the sad adaptor, the PSE-shift was significantly lower than 0 (t (19) = −3.73, $p < .05$, Cohen's $d = -0.83$), indicating that participants were happier with the test images after the sad adaptors than those after the neutral adaptors. Moreover, we used the independent sample t test to compare the emotional adaptation aftereffects between Experiment 1A and Experiment 1B. We found that there were significant differences in the happy adaptor (t (38) = 3.77, $p < .001$, Cohen's d = 1.19), while the differences in sad adaptor were not statistically significant (t (38) = 0.06, $p = .95$, Cohen's $d = 0.02$). The above results replicated the emotional adaptation aftereffect found in Experiment 1B and in previous studies (Wang et al., 2017; Webster et al., 2004; Xu et al., 2008; Ying & Xu, 2017). These results suggest that the emotional adaptation aftereffect is not simply caused by the similar physical features between the adapting and test facial images.

After confirming that Chinese participants showed typical emotional adaptation aftereffects, we further explored whether individuals with low and high levels of depressive symptoms showed differences in the emotional adaptation aftereffect in Experiment 2.

Experiment 2

To explore how level of depressive symptoms affected the adaptation aftereffects and whether such effect was emotion-specific, we recruited participants with low and high levels of depressive symptoms from a relatively large sample. They were given the

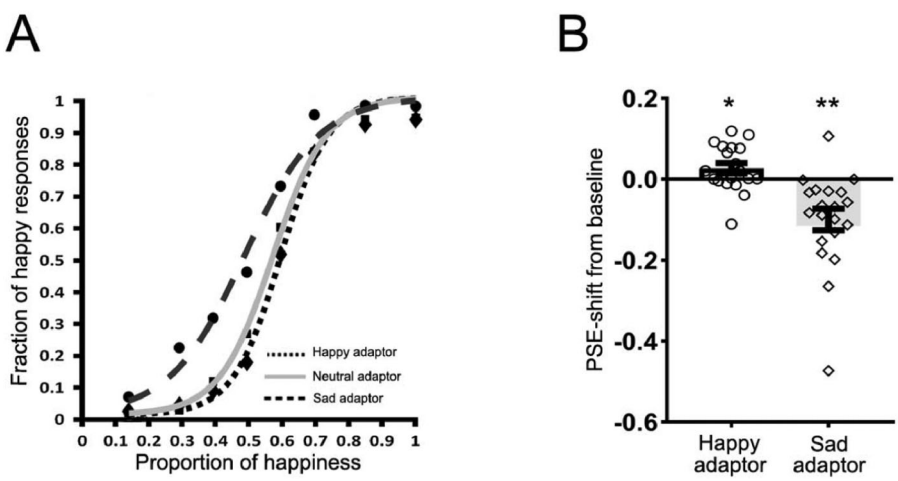

Figure 4. The emotional adaptation aftereffect in Experiment 1B. (A) The fraction of happy responses is plotted as a function of the proportion of happiness of the test images, separately for the happy adaptor condition ($R^2 = 0.96 \pm 0.08$), the neutral adaptor condition ($R^2 = 0.96 \pm 0.04$) and the sad adaptor condition ($R^2 = 0.89 \pm 0.29$); (B) The mean PSE-shift from the baseline condition (i.e. the neutral adaptor condition) for all participants. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. Error bars represent standard errors.

emotional adaptation task as well as a cognitive adaptation task. We hypothesised that the two groups would differ in the emotional adaptation aftereffect but not the cognitive adaptation aftereffect.

Methods

Participants

Initially, 232 volunteers were asked to complete an online questionnaire that included Self-Rating Depression Scale (SDS) (Zung, 1965), Beck Depression Rating Scale (BDI-II) (Beck et al., 1996), and the Raven's Advanced Progressive Matrices (APM) (Raven, 1990). From these participants, 45 college students (30 females, mean age = 21.22, SD = 2.73) were recruited for Experiment 2 based on their scores on the three scales: The high-depression group ($n = 23$) was screened using the criteria of SDS $\geq$ 53 and BDI-II $\geq$ 15, and the low-depression group ($n = 22$) was screened using the criteria of SDS < 36 and BDI-II < 5. As shown in Table 1, the high-depression group had higher scores in SDS (t (43) = 14.24, $p < .001$, Cohen's $d = 4.18$) and BDI-II (t (43) = 11.11, $p < .001$, Cohen's $d = 3.24$). The two groups were matched in terms of age (t (43) = −1.10, $p = .235$, Cohen's $d = -0.36$), gender (χ^2 (1) = 2.61, $p = .106$, $\Phi = 0.38$), and nonverbal intelligence (t (43) = −0.24, $p = .813$, Cohen's $d = -0.07$). None of the participants in Experiment 2 have participated in Experiment 1. All participants were strongly right-handed (Snyder & Harris, 1993), had normal or corrected-to-normal vision, and reported no history of neurological or psychiatric disorders. All participants provided written informed consent, and this study was approved by the Institutional Review Boards of School of Psychology at South China Normal University.

Materials

Eleven facial images and 10 pairs of dot patches were used in Experiment 2. The facial stimuli used in the emotional adaptation task were the same as in Experiment 1. The dot patches were used in the cognitive adaptation task (Turi et al., 2015). They consisted of two adaptor pairs and eight test pairs. One adaptor pair was used as the left adaptor, which included 80 dots in the left position (twice the standard) and 20 dots in the right position (half the standard). The other adaptor pair was used as the baseline, which included 40 dots per side (the same as the standard). The test pairs were composed of 8 images with dots in the left position equal to 24, 28, 34, 40, 46, 52, 58 and 60. The sum of the dot numbers in the left and right positions equalled to 80. For example, if the number of the dots on the left was 60, that on the right would be 20.

Table 1. Mean scores on depression scales and a nonverbal intelligence test for participants with high and low levels of depression.

Scales	High depression (M ± SD)	Low depression (M ± SD)	t	p
SDS	60.39 ± 8.70	31.31 ± 3.90	14.24	<0.001
BDI	19.60 ± 7.88	0.72 ± 1.13	11.11	<0.001
APM	26.49 ± 3.96	26.77 ± 4.13	−0.24	0.813

Note: SDS = Self-Rating Depression Scale (Zung, 1965); BDI = Beck Depression Rating Scale (Beck et al., 1996); and APM = Raven's Advanced Progressive Matrices (Raven, 1990).

Procedure

The procedure for the emotional adaptation task was the same as in Experiment 1. In this experiment, we used a quantitative task to explore the cognitive adaptation because magnitude processing is one of the most central cognitive processing (Turi et al., 2015) and because this task has been used and validated for the investigation of the cognitive adaptation in previous studies (García-Villamisar et al., 2010; Turi et al., 2015). The cognitive adaptation task included the baseline condition and the left adaptor condition. As in the emotional adaptation task, each condition was performed in a separate run with the order of conditions counterbalanced across participants. The two runs were separated by a 5-min break. Each run began with a 1-s fixation. Then, a 4-s adapting stimulus was presented, followed by a 0.5-s inter-stimulus interval. Finally, a test stimulus was presented for 0.2 s (Figure 5). Participants had to judge which side had more dots by pressing one

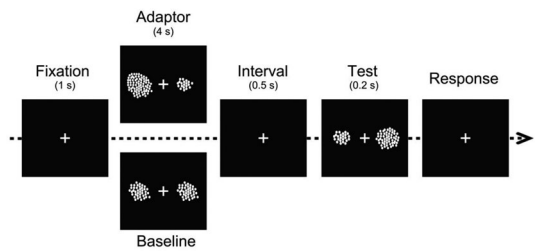

Figure 5. Experimental stimuli and presentation parameters in the cognitive adaptation task. The adapting stimulus consisted of 40 dots on each side in the baseline condition, and 80 dots on the left side and 20 dots on the right side in the adaptation condition. Each adapting stimulus was presented for 4 s, followed by a 0.5-s inter-stimuli interval. The test stimulus was then presented for 0.2 s. Participants were told to judge which side had more dots.

of two keys, and they were asked to gaze at the fixation in the center of the screen. Each test stimulus was randomly repeated for eight times. In total, each run contained 80 trials and lasted for 720 s. Experiment 2 lasted for 60 min.

Data analysis

As in Experiment 1, we calculated the PSEs for the tasks. For the emotional adaptation task, the PSE-shift was computed using the same procedure as in Experiment 1. Those data were then used to explore the differences in the emotional adaptation aftereffect between the low- and high-depression groups. For the cognitive adaptation task, we first calculated the proportion of responses (which side had more dots) to the left side for each test stimulus, and then fitted the data with cumulative Gaussian functions (Fornaciai et al., 2016; Turi et al., 2015), whose median (50% point) was used to estimate the PSE. The PSE-shift was then computed by subtracting the PSE of the baseline condition from that of the left adaptor condition. Finally, a two-way repeated-measures ANOVA and follow-up two-sample *t*-test were used to explore the differences in the emotional and cognitive adaptation aftereffects between low- and high-depression groups.

Results and discussion

The behavioural responses in the three conditions for the two groups of participants were shown in Tables S3–4. As in Experiment 1, we fitted the responses for each participant. The parameter of model fit (i.e. R^2) was 0.91 ± 0.29 in the happy adaptor condition, 0.95 ± 0.20 in the neutral adaptor condition, and 0.92 ± 0.22 in sad adaptor condition. These results suggest that the function fitted the pattern of responses well for all participants.

We then used a two-way repeated-measures ANOVA to explore the differences in emotional adaptation aftereffect (adaptor: happy and sad) across the two groups (depression: high and low). Results showed the main effect of adaptor was significant (F (1, 44) = 199.37, $p < .001$, $\eta^2 = 0.72$). This analysis replicated the results of Experiment 1. Specifically, adapting to the happy adaptor biased the participants' judgment of facial expression toward sadness in comparison to adapting to the neutral adaptor. Conversely, adapting to the sad adaptor biased the participants' judgment of facial expression toward happiness in comparison to adapting to the neutral adaptor (Figure 4A). The adaptor-by-depression interaction was significant (F (1, 44) = 9.57, $p < .001$, $\eta^2 = 0.72$). As shown in Figure 6C, simple effect analysis revealed that, compared with the low-depression group, the high-depression group showed a smaller positive PSE-shift for the happy adaptor condition (F (1, 44) = 9.00, $p < .01$, $\eta^2 = 0.17$), and a smaller negative PSE-shift for the sad adaptor condition (F (1, 44) = 5.24, $p < .05$, $\eta^2 = 0.11$). The main effect of depression was not significant (F (1, 44) = 0.01, $p = .937$, $\eta^2 = 2.37 \times 10^{-5}$). These results indicate that compared

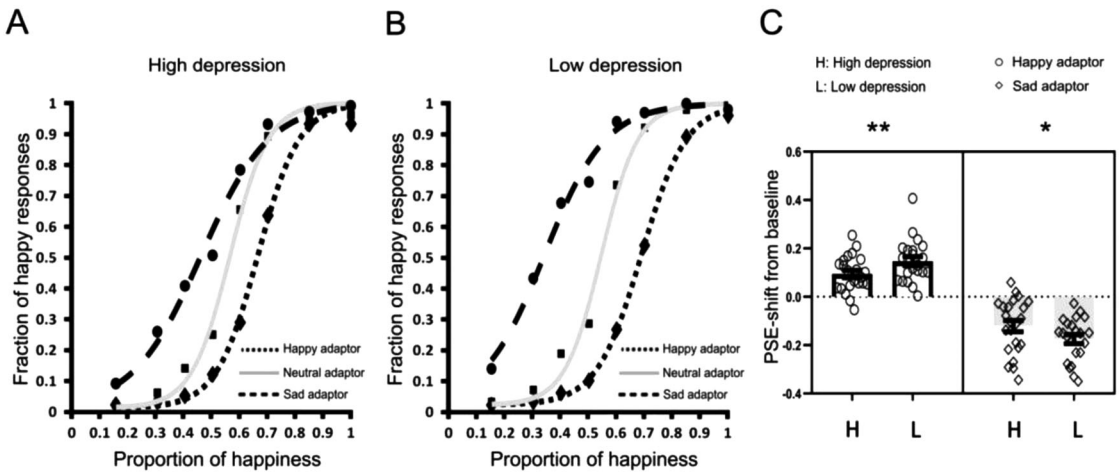

Figure 6. The emotional adaptation aftereffects in Experiment 2. (A) Psychometric functions of the high-depression group (Happy: $R^2 = 0.90 \pm 0.29$; Neutral: $R^2 = 0.92 \pm 0.27$; Sad: $R^2 = 0.88 \pm 0.30$); (B) Psychometric functions of the low-depression group (Happy: $R^2 = 0.91 \pm 0.30$; Neutral: $R^2 = 0.98 \pm 0.03$; Sad: $R^2 = 0.95 \pm 0.05$) and (C) the differences in PSE-shift from the baseline condition (i.e. the neutral adaptor condition) between the two groups. Error bars represent standard errors. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$).

to individuals with low depression symptoms, those with high depression symptoms were less affected by the happy and sad adaptors.

The behavioural responses in cognitive adaptation task were shown in Tables S5–6. We then fitted the responses for each participant. The parameter of model fit (i.e. R^2) was 0.98 ± 0.03 in the adaptor condition, and 0.97 ± 0.03 in baseline condition. These results suggest that the function fitted the pattern of responses well for all participants. To examine whether the between-group differences in the adaptation aftereffect were emotion-specific, we compared the cognitive adaptation aftereffects between the two groups. In this analysis, we first performed a one-sample *t*-test on the data of all participants and found that the cognitive adaptation aftereffect (i.e. PSE-shift) was significant (t (45) = −16.47, $p < .001$, Cohen's $d = -2.45$). We then conducted a two-sample *t* test and found that the cognitive adaptation aftereffects did not differ between the two groups (t (44) = 0.38, $p = .717$, Cohen's $d = 0.11$) (Figure 7C). These results suggest that participants' level of depressive symptoms have no significant impact on the cognitive adaptation aftereffect.

It should be noted that, following previous studies (Fornaciai et al., 2016; Turi et al., 2015), we used a different function to model the data of the cognitive adaptation task from the emotional adaptation task. To exclude the possibility that the different findings for the two tasks were caused by the use of different functions, we further fitted the data from the cognitive adaptation task using the same function as in the emotional adaptation task ($f(x) = 1/(1 + e^{-a(x-b)})$). Results showed that the cognitive adaptation aftereffects still did not differ between the two groups (t (44) = 0.38, $p = .717$, Cohen's $d = 0.11$) (Figure S1). These results suggest that the differential results for the two tasks were not caused by the use of different functions.

General discussion

Using the adaptation paradigm, the present study aimed to investigate the relationship between participants' level of depressive symptoms and adaptation aftereffects. Experiment 1 confirmed that Chinese participants showed the typical emotional adaptation aftereffect. More importantly, Experiment 2 showed that participants' level of depressive symptoms was associated with the emotional adaptation aftereffect, but not with the cognitive adaptation aftereffect. Specifically, compared to the low-depression group, the high-depression group showed a smaller emotional adaptation aftereffect in both happy and sad adaptor conditions. These results suggest that the level of depressive symptoms affected the adaptation aftereffects and such effect was emotion-specific.

Experiment 1 replicated the emotional adaptation aftereffect (Webster et al., 2004) in Chinese participants. Our results together with previous findings in Singaporean and American participants (Hong & Yoon, 2018; Sou & Xu, 2019; Wang et al., 2017; Ying & Xu, 2017) suggest that the emotional

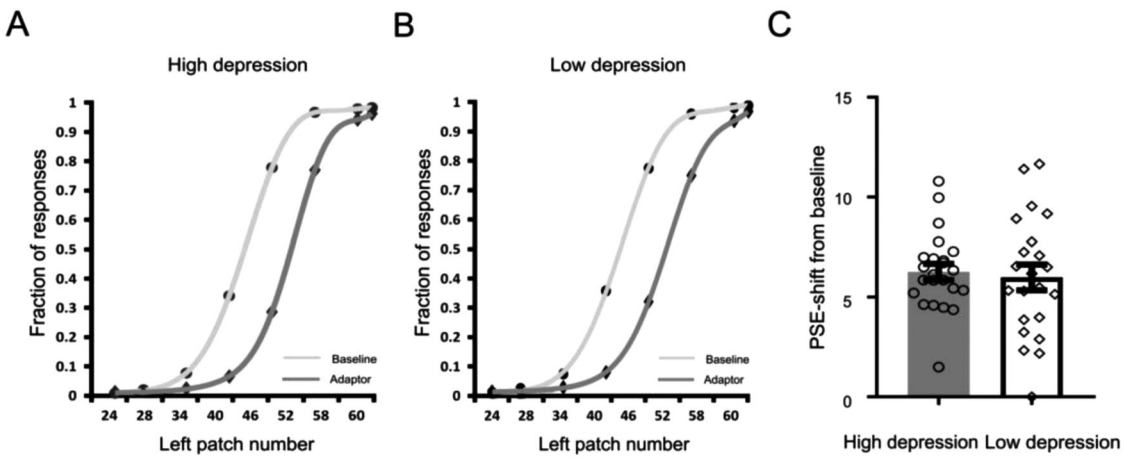

Figure 7. The cognitive adaptation aftereffects in Experiment 2. For the psychometric functions, the proportion of trials in which participants reported the left side having more dots is plotted as a function of the actual number of dots on the left side. The psychometric functions are plotted for participants with high level of depressive symptoms (A) (Adaptor: $R^2 = 0.99 \pm 0.03$; Baseline: $R^2 = 0.97 \pm 0.03$), and those with low level of depressive symptoms (B) (Adaptor: R^2: 0.98 ± 0.02; Baseline: R^2: 0.97 ± 0.02). Panel C shows the mean PSE-shift from the baseline condition for the two groups with different levels of depression. Error bars represent standard errors.

adaptation aftereffect could be stably produced in different populations. Our study further revealed that the emotional adaptation aftereffect was not simply induced by the similarity in physical features between the adapting and test stimuli, but by the emotional information. Consistent with our findings, the emotional adaptation aftereffect has been suggested to reflect the process of sensory adaptation at a higher level than perceptual judgment (de la Rosa et al., 2013; Leopold et al., 2005; Smortchkova, 2020), perhaps because the emotional adaptation coding reflects not only early sensory processing but also the higher-level coding of facial features and facial expression (Butler et al., 2008; Palermo et al., 2011; Wallraven et al., 2005; Yovel & Duchaine, 2006). These findings consistently indicate that emotional adaptation aftereffect is a stable advanced adaption effect.

The occurrence of the emotional adaptation aftereffect could be explained by affective counter-regulation, which has been assumed to reflect a highly adaptive and automatic mechanism underlying emotion regulation (Rothermund, 2011; Rothermund et al., 2008; Schwager & Rothermund, 2013, 2014; Wentura et al., 2018). According to the counter-regulation principle, attention is preferred to be allocated to information that is opposite in valence to the current emotional or motivational state. Specifically, a positive emotion should promote automatic allocation of attention to negative stimuli, whereas a negative emotional state should promote automatic allocation of attention to positive stimuli. Such a mechanism enables the motivational and emotional homoeostasis. Consistent with this view, previous research found that individual's responses to the target stimuli were influenced by emotional conditions (Schwager & Rothermund, 2013). From this perspective, the emotional adaptation aftereffect found in this study might reflect the affective counter-regulation from the adaptor to the test stimulus.

More importantly, our study revealed a smaller emotional adaptation aftereffect for the high-depression group than for the low-depression group, perhaps as a result of a difference in the affective counter-regulation between the two groups. As discussed above, counter-regulation plays an important role for the automatic regulation of emotions themselves. It helps to regulate emotional states and to restore a balanced sensitivity for positive and negative information (Rothermund, 2011; Rothermund et al., 2008; Rothermund et al., 2011; Schwager & Rothermund, 2013, 2014). According to this functional view, a smaller emotional adaptation aftereffect for the high-depression group may reflect the result of a break-down in adaptive affective counter-regulation. For the high-depression individuals, affective attentional allocation has lost its flexibility and no longer serves the function of preventing an escalation or perseveration of affective states. Consistent with this view, there is psychopathological evidence that individuals with symptoms like anxiety or depression tend to show rigid affective processing biases (Bar-Haim et al., 2007). An alternative explanation for group differences in the emotional adaptation aftereffect is the emotional appraisal theory. According to this view, the interpretation/appraisal of an emotional situation is continually modified by reappraisals over time (Ellsworth & Scherer, 2003; Siemer et al., 2007). Such a modification process, however, is less effective in individuals with depression (Sheppes & Meiran, 2007; Smith et al., 2016). Based on such evidence, we infer that the high-depression individuals in our study might have a maladaptive emotional reappraisal system, so they were less likely to be influenced by the happy and sad adaptors, more likely to maintain their original judgments (i.e. response under the neural adaptor condition) on the test images, and hence showed a smaller emotional adaptation aftereffect. Since our study is not able to differentiate the two explanations, following up on the contributions of counter-regulation and reappraisal to the emotion adaptation aftereffect seems an interesting avenue for future research on emotional adaptation.

By including a control task (a cognitive adaptation task), our results suggest that the effect of participants' level of depressive symptoms on the adaptation aftereffects is specific to emotional stimuli. These results are consistent with several studies showing that depressed individuals and healthy controls differ in emotional processing, but not in cognitive processing (Fales et al., 2008; Fales et al., 2010). Indeed, emotional and cognitive processing has been found to rely on different neural networks (Pessoa, 2017; Zinchenko et al., 2017). Consequently, individuals with cognitive impairments (e.g. learning disabilities) do not necessarily have emotional disorders (Gajre et al., 2016; Schwab, 2014), and some individuals with emotional disorders perform cognitively well (Fales et al., 2010; Manning et al., 2019). The specific link between depression and the emotional adaptation aftereffect has important implications for the diagnosis of depression. As a smaller cognitive adaptation aftereffect has been used to identify cognitive disorders such as dyslexia or other

learning disabilities and even autism (García-Villamisar et al., 2010; Perrachione et al., 2016), a smaller emotional adaptation aftereffect may be used as an indicator of depression or related depression tendency.

A major limitation of our study should be addressed in future research. In this study, we recruited college students with high and low levels of depressive symptoms as participants, but they were not officially diagnosed by clinicians. Future research should specifically recruit a clinical sample of individuals with major depression disorder to investigate their emotional adaptation aftereffect as compared to healthy controls. We are confident of our results, however, because there is evidence that individuals having a high level of depressive symptoms following the procedures we used showing similar behavioural performance in a variety of emotional operations as individuals diagnosed as major depression disorder (Geerts & Bouhuys, 1998; Huprich et al., 2012; Vittengl et al., 2014).

In summary, the present study used the adaptation paradigm to reveal that participants' level of depressive symptoms was associated with the emotional adaptation aftereffect. It provided a potential indicator (i.e. the emotional adaptation aftereffect) for distinguishing individuals with high and low depressive symptoms.

Author contributions

Nan Jiang: Software, Formal analysis, Writing-Original draft preparation, Visualization, Writing-Reviewing and Editing. **Huiling Li**: Data curation, Writing-Original Draft, Writing-Reviewing and Editing. **Chuansheng Chen**: Supervision, Writing-Reviewing and Editing. **Ruilin Fu**: Investigation, Visualization. **Yuzhou Zhang**: Formal analysis, Visualization. **Leilei Mei**: Conceptualization, Methodology, Writing-Original draft preparation, Writing- Reviewing and Editing, Supervision, Funding acquisition.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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